

1. Which one of the following estimates is unrealistic?

- A The potential energy of a man at the top of Bukit Timah Hill is 100 kJ.
- B The volume of air in a classroom is 150 m³.
- C The mass of an apple is 100 g
- D The average temperature of fire is 300 K.

Ans: D

A: Height of Bukit Timah Hill = 160 m,
potential energy of man = $mgh = (60)(9.81)(160)$
= 94 kJ

B: Volume of air in classroom = volume of classroom
= 7 x 8 x 3
= 147 m³

C: 100 g for an apple is quite reasonable right...

D: 300 K is 27 °C, so is unrealistic for the temperature for fire.

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2 The velocity- time graph of a particle moving in a straight line is shown in Fig. 2: